Shi Hao Ng

+44-7436-514602 | shi-hao.ng25@imperial.ac.uk | linkedin.com/in/shi-hao-ng | github.com/Ice-Citron | sienarindustries.com

Education

Transfer to BEng requested; expected graduation Jun. 2028

Imperial College London — London, UK
Master of Engineering: Computing AI/ML, Year 1: Upper Second Class (2:1) — *Sep. 2025 – Jun. 2029*

Marlborough College Malaysia — Johor, Malaysia
IB Bilingual Diploma: 42/45 (7, 7, 7 in Higher Level subjects) — *Graduated May 2025*

- IGCSE: 10A* 2A. Best score in Asia for Computer Science; Best in Malaysia for Design Technology, Physics & Chemistry

Experience

Robotics & Machine Learning Intern — Jul. 2026 – Sep. 2026
Vikaso Robotics 4.0 — *Buckinghamshire, UK*

- Built Vikaso's first camera-intrinsic and camera-to-robot extrinsic calibration pipeline, achieving ∼3 mm initial extrinsic accuracy; deployed and tested for 100+ hours on Universal Robots UR5e arm & Intel RealSense camera
- Built Vikaso's first configurable perception stack, integrating 3 segmentation models and 4 CAD-based 6D pose-estimation models (all open-source) behind a common interface that accepts RGB or RGB-D input and returns masks and 6D poses
- Built a benchmark harness evaluating integrated models on 6D localization and 2D detection tasks across 10 public datasets. Results matched each model's published Average Recall, which verified the integrations reproduced the papers

Re-Founder & Co-President — Oct. 2025 – Apr. 2026
Imperial College Drone Society — *London, UK*

- Re-established Imperial College Drone Society, secured £3.8k sponsorship pledges, 80+ members, ran drone-building events

Engineering Intern — Jun. 2025 – Jul. 2025
Nosco Asia (S) Pte Ltd — *Taichung, Taiwan*

- Assisted hot-repair ceramic-welding crews servicing still-running large-scale glass furnaces. Operated welding machinery, performed ceramic welding, cut refractory bricks, and learned how glass production lines operate

Projects

Intrinsic AI for Industry Challenge | *PyTorch, LeRobot ACT, Isaac Sim, Gazebo* | Website — 2026

- Experimented with RL (Isaac Lab), imitation learning (ACT) and scripted FSM policies for fiber-optic cable insertion into server-rack ports on a simulated UR5e; placed 49th of 196 teams with Team Quantarune
- For the final submission, developed a hybrid ACT–FSM policy using ACT (Action Chunking Transformers) for approach and cable-to-port alignment, followed by rule-based insertion with force-scaled steps and lateral motion limits. Added automatic retraction and realignment to recover from failed attempts
- Scored 89 on held-out competition evals versus 141 locally. Later analysis found the local tests did not reflect competition conditions, motivating a paper on the policy (in preparation)

GPT-2 Pre-training: Layer Normalization Study | *PyTorch, DDP, HPC* | GitHub | Paper — 2024 – 2025

- Reproduced GPT-2 124M from scratch in PyTorch, pre-trained on FineWeb using DDP on 4× NVIDIA H100s; tuned GPU configurations, batch size, mixed precision and `torch.compile`, achieving training throughput of 1.27M tokens/s
- Implemented LayerNorm, RMSNorm & PowerNorm. RMSNorm achieved comparable loss at 6% higher throughput than LayerNorm; PowerNorm's gradients exploded. Ran normalization ablations and analyzed per-layer gradients
- Fine-tuned base & ablated models on BillSum and SQuAD. Evaluated summarization and QA using ROUGE/F1 and GPT-4o (LLM-as-a-judge). Published weights on Hugging Face; awarded Grade A for this IB Computer Science Extended Essay

Google EDUtech Asia: Geospatial Waste Analytics | *YOLOv11, PyTorch, Fine-tuning, GCP* | Website | GitHub — 2024

- ML Lead: Won 1st place and keynoted to 1k+ people. Built waste analytics platform that maps trash distribution: drone images are sent to fine-tuned YOLOv11-L, storing detections in Flask/Firebase backend and plotting heatmap with Google Maps API
- Collated 300k images from 25 waste datasets and merged 31 label names into 8 classes. Fine-tuned YOLOv11-L for 9 epochs on GCP's NVIDIA H100, reaching 0.381 mAP50 and 0.264 mAP50-95 on eval set; inference runs at ∼15 ms/image on NVIDIA L4

C++ OpenGL Game Engine | *C++17, OpenGL, GLFW, ImGui* | GitHub | Website — 2022 – 2023

- Recreated game engine in C++/OpenGL following The Cherno's tutorials: batched sprite renderer with texture slots over shared buffers, GLSL shaders for texturing and lighting, and a hand-written matrix library with a transformation stack
- Built an application framework using GLFW for windows & input and C++ templates to dispatch events to handlers, with a layer stack that routes events to overlays first. Integrated ImGui docking and multi-viewport support into the render loop

Augmented Electromagnetic Railgun | *LS-Dyna, Fusion 360, High-Voltage Electronics* | Website | Paper — 2024 – 2025

- Iterated through 3 physical prototypes modeled in Fusion 360; ran 70 LS-DYNA simulations across 10 voltage conditions to settle on an augmented railgun design. Built a 450 V, 10 mF capacitor bank for the power system; final railgun averaged 132 km/h across 10+ firing trials, estimated from high-speed footage

Custom FPV Racing Drone | *Blender, Betaflight, PET-CF 3D Printing, ELRS* | Website — 2025

- Designed the frame in Blender and 3D-printed in PET-CF; hand-soldered electronics (Mamba F722, DJI O4 Pro, ELRS), tuned Betaflight PIDs. Replaced frame due to crash damage; flew the completed 610 g drone in Acro (240 km/h theoretical top speed)

Technical Skills

Languages: Python, C/C++, Kotlin, Java, Haskell, TypeScript/JavaScript, CUDA, RISC-V assembly
ML: PyTorch, DDP, Hugging Face Transformers, LeRobot, Weights & Biases, NumPy, vLLM, llama.cpp
Computer Vision: OpenCV, YOLO, SAM, CNOS, FoundationPose, MegaPose, 6D pose estimation, camera calibration
Robotics & Simulation: ROS 2, MuJoCo C API, Gazebo, Isaac Sim, Universal Robots UR5e, NVIDIA Cosmos 3
Systems & Web: Linux/CUDA toolchain, Git, Docker, GCP, AWS, OpenGL, GLFW, React, Node.js, Firebase
FEA & Hardware: Fusion 360, Blender, Ansys Fluent, LS-Dyna, 3D printing, CNC, high-voltage electronics, Betaflight, soldering